



Source-Native Benchmarking for Nature-Inspired Time-Series Families: A Fail-Closed Experimental Protocol

LumenCore Technical Note - Current Public-Safe Review Draft

This technical note reports local software, custody, benchmark, and protocol states. It is not peer review, independent validation, a performance claim, field validation, trading alpha, realized savings, enterprise value, a patent opinion, or deployment authority.

Abstract

LumenCore registers candidate computational families inspired by natural forms, but does not treat inspiration as evidence. This note reports a source-native benchmark ledger, the retrospective disposition of prior leads, and a frozen prospective protocol. The current ledger contains 140 registered families, 35 implementations, and 126 direct candidate-source-baseline comparisons. No candidate passes the promotion gate. The scientific contribution is therefore a reproducible comparison and falsification framework, not a performance champion.

Research Question

Can a predeclared candidate family beat every accepted baseline for a specific source, series, cadence, and horizon under prospective custody, clustered inference, an effect floor, and familywise error control?

Current Evidence Snapshot

State	Count
Registered families	140
Implementations present	35
Missing implementations	105
Direct candidate-source cards	23
Direct comparisons	126
Global Holm-positive comparisons	0
Promoted champions	0
Market-signal comparisons	48
Market comparisons inferentially insufficient	48
Prospective eligible observations	0

1. Source-Native Experimental Design

Why source-native baselines matter

A candidate can appear strong when it is compared against an inappropriate cadence, horizon, seasonal period, or baseline. This protocol therefore treats the source contract as part of the hypothesis. A family must win within each named source against the complete accepted baseline roster before any broader promotion.

Evaluation controls

- Candidate-by-source cards evaluated against the baseline roster registered for that source, cadence, series, and horizon.
- Expanding history ends immediately before each forecast origin.
- Inference clusters overlapping origins and horizons at the source-series level instead of treating every row as independent.
- Global Holm correction across the candidate-source-baseline comparison family; prior subset leads receive no preference.
- Positive, neutral, negative, inconclusive, and invalid outcomes are retained.

Registered baseline roster

- naive_last	- drift
- moving_average	- exponential_smoothing
- linear_trend	- seasonal_naive_source_period
- damped_holt_ets	- autoregressive_ridge_source_lag

Retrospective result

The former FRED and TWELVE_DATA subset leads are retired because they do not survive source-series clustering and the complete source-native baseline gauntlet. Four fixed market-signal families were also run against four registered baselines on each of Kraken, TwelveData, and AlphaVantage. All 48 market comparisons are inferentially insufficient because each source currently supplies one source-series cluster. Across the ledger, zero comparisons survive the global correction and zero candidate-source cards pass promotion.

Current conclusion: no alpha, edge, or champion is established.

2. Frozen Prospective Protocol

Candidate identity

The registry label is **fractal_brownian_surface**. The scientific estimator is **hurst_conditioned_multiscale_increment_heuristic_v1**: Hurst-conditioned weighted multiscale increment heuristic using lags 1, 2, 4, and 8 with a bounded persistence multiplier. The estimator name prevents the registry label from being mistaken for a validated fractional Brownian motion model.

Primary endpoint and multiplicity

The endpoint is `relative_mean_absolute_error` ($rMAE = MAE_candidate / MAE_baseline$). The candidate must clear 16 predeclared contrasts under `one_sided_holm_familywise` at familywise alpha 0.05. The maximum accepted rMAE is 0.95, corresponding to at least 5 percent relative improvement.

Future-only sample gates

- Daily arm: at least 520 future eligible sessions per series and 104 nonoverlapping five-session blocks.
- Monthly arm: at least 60 future releases per series.
- Uncertainty: `synchronized_moving_block_bootstrap` with 20000 frozen replications.
- Sample gates cannot be reduced after freeze.

Current protocol state

Protocol	State
LUMENCORE_TS_SOURCE_NATIVE_20260729_V1	FROZEN_AWAITING_FUTURE_OBSERVATIONS
Promotion decision	WAITING_FOR_NEW_SOURCE_ROWS
Eligible future observations	0

Decision states

- **replicated_lead**: All three former lead contrasts must pass inside the full 16-test Holm family and each adjusted one-sided bound must clear the 5 percent effect floor.
- **source_winner**: The candidate must beat all eight source baselines, no series-horizon cell may be worse by more than 5 percent, and the p95 error ratio must not exceed 1.10.
- **cross_source_promotion**: A source winner must repeat in a second independent prospective period.
- **integrity_failure**: INVALID
- **insufficient_sample**: INCONCLUSIVE_WAITING_FOR_NEW_SOURCE_ROWS
- **primary_failure**: CORRESPONDING_CLAIM_FALSIFIED

3. Contribution, Limits, and Reproducibility

Scientific contribution

- A source-native baseline contract that prevents cross-source or cadence-mismatched promotion.
- A custody gate that binds exact source snapshots before benchmark acceptance.
- A clustered inference rule that avoids pseudoreplication from overlapping forecast origins and horizons.
- A full-family multiple-testing rule that prevents cherry-picking isolated wins.
- A future-only protocol with fixed endpoints, effect floor, sample gates, ablations, and falsification states.
- A cost-aware market-signal replay that uses identical timestamps, future-return rows, turnover costs, and source-specific baselines without granting a promotion from descriptive wins.
- A machine-readable claim boundary that keeps software proof separate from field, economic, or deployment claims.

Limitations

- 105 of 140 registered families lack implementations.
- Only 3 lanes currently have executable direct measured adapters; the wider nature-inspired registry remains inventory, synthetic stress, or context until implemented.
- All 48 market-signal comparisons are inferentially insufficient under the predeclared five-cluster minimum; descriptive score differences are not alpha or edge.
- The prospective protocol has zero eligible future observations and cannot yet support a prospective accuracy conclusion.
- No result establishes universal superiority, field performance, trading alpha, realized savings, customer acceptance, or deployment authority.

Reproducibility receipts

The ledger, prospective configuration, and prospective status are bound below. Hashes establish byte identity and custody only; they do not establish scientific correctness or external acceptance.

Artifact	Bytes	SHA-256
out/ops/source_native_family_baseline_ledger_latest.json	2159316	7bc80126d77e8edec39d0fe97adff15c477ba681a985b0dabb431c0b4e963d2d
config/time_series_source_native_prospective_protocol_v1.json	7735	bec90c816a08ce493c5113ffd8ee6a3f48348663ab73f1f6e484d7c2199acbcd
out/ops/time_series_source_native_prospective_protocol_status.json	1108	b66af0ca033d95982b4b38c2ca9699a1386b3addf07a56b0c988c73125c508cd

Legacy concept-paper disposition

Three older papers are preserved as historical speculative concepts and blocked from upload. Their BioGeometry, scalar-field, bioresonance, cooling-savings, zero-point, weather-control, and wormhole-adjacent claims are not supported by the current source-native benchmark evidence.

Whitepaper payload SHA-256: 6d7ba7e4d1d119c0755fa15bb22fa91c39d0220caccb21fe5f1b69c24165e706